

Title: State Capacity, Political Goods, and Energy Transitions: A Comparative Analysis of SDG 7 in Bangladesh and Japan

Course: POL 102: Comparative Governance (Section 1)

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Introduction

Sustainable Development Goal 7 (SDG 7) establishes a critical international mandate: Access to affordable, reliable, sustainable and modern energy for all populations by 2030. This goal is crucial not only for reducing the environmental impact of climate change but also for eradicating systemic poverty, fostering economic productivity, and improving human dignity in both the developing and developed world (United Nations, 2023). However, establishing global goals is very different from attaining them. It is the underlying administrative capacity of a sovereign state that produces the complex public goods of energy under different historical, economic and institutional conditions that is the key variable that turns international energy goals into domestic policy products.

This paper compares energy transition trajectories of Japan and Bangladesh to systematically analyze these dynamics. The theoretical approach of Institutionalism and Political-Economy are used to understand how these two countries, with very different approaches to energy provision, deal with the modern challenges. In methodological aspect, this paper is based on Most Different Systems Design (MDSD). The MDSD approach can isolate the impact of state capacity on the delivery of public services, even in the face of very different economic starting-points, by comparing a highly developed, post-industrial democracy (Japan) with a post-colonial, rapidly developing Lower Middle-Income Country (Bangladesh).

The central and comparative puzzle underlying this research is that Bangladesh and Japan have such highly asymmetrical SDG 7 outcomes – despite both states committing to sustainable development. To address this, the study conducts a longitudinal analysis of the three indicators—electricity access, clean cooking access, and renewable energy consumption—from 2000 to 2021. This empirical evidence shows an intriguing split. In this period, although Japan enjoyed universal access to basic services, the country's sustainable development of renewable energy was far behind that of other comparable advanced economies. In contrast, Bangladesh achieved impressive infrastructural development, lifting electricity access from about 32 per cent in 2000 to almost 100 per cent in 2021, thereby building a solid model for other LMICs. However, its share of renewable energy fell sharply and its access to clean cooking remained stagnant at 28 percent. This paper sheds light on the institutional, political and structural constraints that define global energy transitions by unpacking these contrasting trajectories and the survival strategies of those that have been adopted.

Literature Review

For understanding, however, the dynamic structures which drive state action and shape the outcomes of policy at the level of the state, comparative governance research is structurally required. As Lijphart (1971) has famously suggested the comparative method is an important addition to the experimental method in political science because it allows the researcher to establish the empirical relationship between variables by isolating the systemic similarities and differences. Based on this methodology, Geddes (2003) emphasizes that comparative research design must be strong, by attentively selecting the cases and by application of the empirical evidence rigorously testing the theoretical paradigms, avoiding the danger of selection bias. If the critical question on the delivery of a complex public service such as energy is whether formal institutional designs or material constraints matter most in order to explain the success of policy decisions, then it is useful to compare countries with radically different starting points of their economies.

However, development and capacity of the state is not a vacuum, but strongly shaped by global hierarchies and past experiences. According to Wallerstein (1974) the international political economy provides the mechanism through which financial resources, technological changes and policy paradigms move from one country/region to another. It is therefore important to recognize that the state formation in the past ends up shaping the administrative capacity in the present as presented in the example of a highly developed core nation such as Japan and a post-colonial, peripheral country like Bangladesh (Lange, 2012).

This history is intimately related to the theoretical spectrum strong and weak states. Skocpol (1985) made the famous claim to 'bring the state back in' to political analysis, and state capacity as he sees it is the ability of a state to set up and pursue official goals with autonomy. Rotberg (2003) carries this a step further, arguing that a state's most basic political good is to ensure human and territorial security: without it, the state cannot deliver the higher order goods of infrastructure. Japan is a classic strong state: high autonomy and absolute provision of security allow deliverables to be executed frictionlessly, with hierarchy. Instead, Bangladesh is on the low end of that spectrum. Although it has made tremendous macro-economic progress, it has delivered decentralised services asymmetrically and remains highly vulnerable to political and

institutional shocks, as evidenced by its continued ranking on the Fragile States Index (Fund for Peace, 2024).

An Institutionalist approach to theory helps to understand these divergent SDG 7 outcomes systematically. Acemoglu and Robinson (2012) argue that the key reason for the provision of sustainable public goods is the presence of inclusive political and economic institutions, while extractive institutions focus on the short-term centralization of power. So institutionalism is concerned with these “rules of the game”, as both formal and informal rules can be important drivers of outcomes of policies because of the role played by bureaucratic path dependency (McCormick et al., 2019). Decades of institutional stability in Japan have created a lot of bureaucratic inertia. The state monopoly has pushed back hard against the rapid and disruptive processes of institutional change needed to get the country's power grid decentralised and onto renewables. Against this backdrop, the top-down power sector institutions in Bangladesh have demonstrated enormous success in pursuing top-down expansion of the grid – the "hardware" of development – but remain woefully under-resourced when it comes to what can be considered the regulatory "software" of community-level transitions, such as clean cooking (Sovacool et al., 2020).

In addition, distribution of these public goods is strongly shaped by theories of nationhood, social cohesion and democratic consolidation. As Brubaker (1992) and Anderson (1983) indicated, the way a State imagines its "imagined community" will be the key to determining its ability to make equitable distributions. The near homogeneity of both of Japan's civic and ethnic groups helps to assure an even, universal distribution of public services, whether in rural or urban areas. By contrast, developing states with polarized societies often are unable to distribute resources fairly (Posner, 2004; Varshney, 2001). This polarization can become very pronounced if there is challenge to the basic political equality (Dahl, 1977), or if executive power takes over in the sensitive process of transitioning to democracy (Isumona, 2012; Schmitter & Karl, 1991).

Ultimately, these infrastructural choices are inextricably linked to regime survival. Populist strategies work very much on the speed and the visibility of the delivery of mass infrastructural goods to cement the argument that, exclusively, the regime is dedicated to providing services to the people (Müller 2016). This is, therefore, a highly politicised and economic calculation of energy policy. Traditionally, Japan's so-called "Iron Triangle"—conservative politicians, state

bureaucrats and industrial utilities conglomerates—has been strong at stifling renewable energy momentum, in order to preserve vested economic interests in fossil fuels. Bangladeshi foreign donors are shaping political economy through huge import of Liquefied Natural Gas (LNG) and coal; state is gaining domestic political legitimacy through rapid electrification.

Empirical Evidence and Longitudinal Trajectories

Existing research in comparative governance heavily depends on homogenized and aggregated data to monitor sustainable development and policy change in themes between countries. The seminal work by leading experts on state effectiveness, like Kaufmann, Kraay and Mastruzzi (2010), determined that empirical measures of governance are necessary for cross-national assessment. Moreover, energy economists often resort to the World Bank Data for an empirical study on the impact of electrification programmes in developing countries as is the case of Bhattacharyya (2012). In this regard, the study builds from the world-known academic methodology and uses the World Development Indicators (WDI) dataset from the World Bank, covering the time period 2000–2021 (World Bank, 2025). The data is organized around three indicators that represent different aspects of state capacity: (1) basic access (hardware); (2) localized clean technology (software); and (3) sustainable transitions.

Energy access as measured by electricity (National Electricity Access (% of Population))

This indicator is the percentage of the population of a country with basic access to electricity (column B), ranging from 0 (low) to 100 (high). The higher score indicates a robust institutional ability to provide large-scale bottom-up infrastructures/ 'hardware', with significant capital required and central management of the grid.

As the data shows (Appendix, Table 1), Japan had a perfect electricity access rate of 100% for the entire period of 2000 to 2021. This is testimony to its very centralized electric energy sector, formed in the reconstruction of the post-WWII era, which has been good in guaranteeing the provision of political basic goods. On the other hand, Bangladesh shows an impressive and almost unparalleled upward trend with its access rate increasing from about 32% in 2000 to more than 99% in 2021.

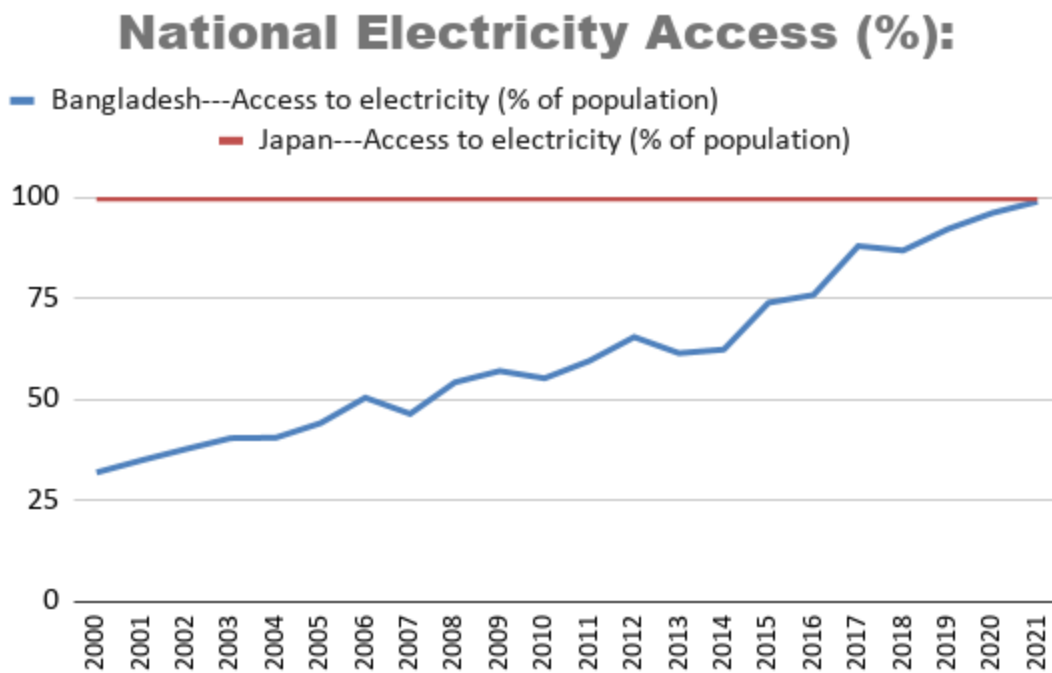


Figure 1: National Electricity Access (% of Population) from 2000 to 2021. Data Source: World Bank.

The key comparative question about this trajectory is: What were the institutional mechanisms for, and why, this rapid progress? The real-world context would explain that this pressing need for electrification was one that was driven, in part, by a deep and dangerous crisis of domestic political legitimacy. The Caretaker Government (CTG) system was abolished in 2011, which was controversial and repeatedly challenged the ruling government for electoral fairness and democratic legitimacy (Khan, 2023). The government was in dire need of "output legitimacy" (visible and tangible economic deliverables) to bring in massive support from the voters and avoid civil unrest in the absence of "input legitimacy" (fair and broadly accepted elections).

The state did this in a very effective way and by using two approaches. First, it was a big subsidy for immediate, highly visible, grid expansion, through the authorization of Quick Rental Power Plants (QRPPs). These QRPPs circumvented the traditional and time-consuming public procurement processes, delivering quick "Electric" hardware—with a heavy dependence on fossil fuels. Second, acknowledging the shortcomings of the central grid for remote locations, the state encouraged widespread development of off-grid solar energy through "Solar" Home

Systems (SHS) in rural communities. This combination of a top down fossil-fuel grid and local solar distribution is a key enabler for Bangladesh's rapid success in electrification and provides a highly relevant, empirical demonstration of a model that could be replicated for other Lower Middle-Income Countries (LMICs) with similar challenges of high energy poverty.

Access to Clean Fuels and Technologies for Cooking (% of Population)

This indicator represents the proportion of the population using cleaner cooking technologies (electricity, LPG, natural gas, etc.) as opposed to highly polluting traditional biomass sources (wood, charcoal, dung, etc.). It is not a grid expansion metric like above, but measures a state's capacity to make localized implementation of regulatory "software" possible, which demands extensive engagement within communities, poverty alleviation, and the capacity to cause behavior change at the household level (Pachauri et al., 2013).

The overall trend for Japan's score is a flat line at 100% (see Appendix, Table 2), as Japan has a regulatory state at a very high level of development that can easily enforce health and safety at the town level. But the story of Bangladesh is starkly different and alarming, relating to its past achievements in the electrification arena. Although the general electricity access has seen a huge jump, clean cooking simply ticked over from 7.7% in 2000 to around 25.9% in 2021.

Clean Cooking Access (%): Bangladesh vs. Japan (2000–2021)

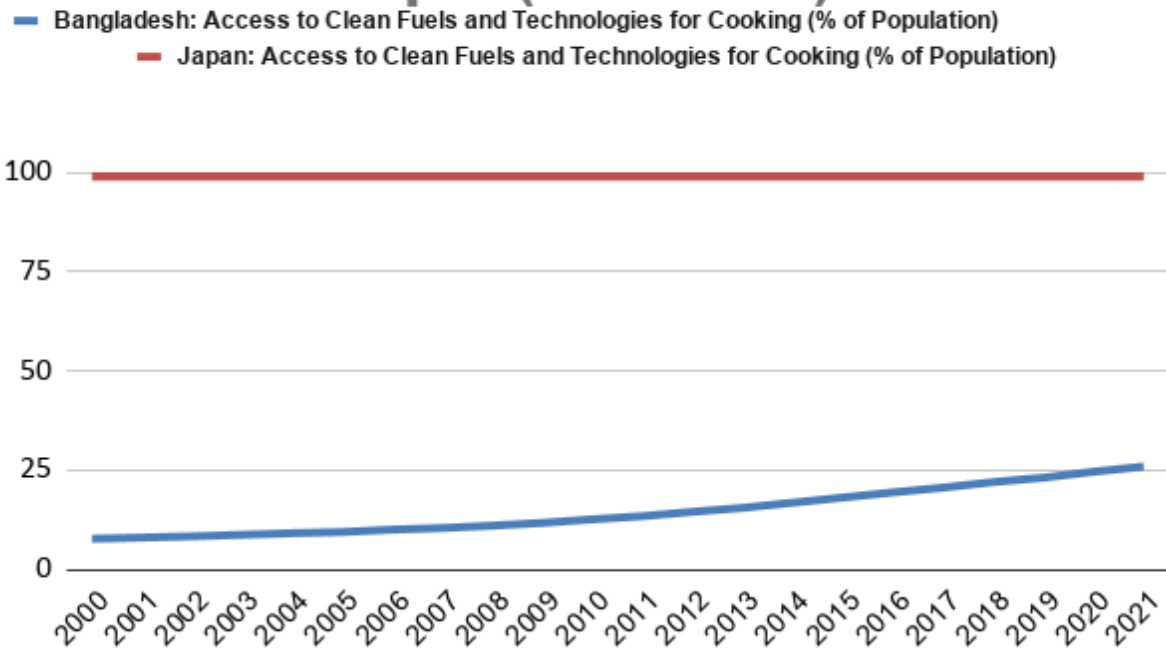


Figure 2: Access to Clean Fuels and Technologies for Cooking (% of Population) from 2000 to 2021. Data Source: World Bank.

One of the key inflection points was after 2010, when electricity grew rapidly, and clean cooking trends did not track the explosive growth of electricity access towards political capital. This empirical fact reflects a deep asymmetry in state capacity. The central government in Dhaka has the ability to procure highly visible infrastructural hardware (power plants and transmission lines) with all efficiency for quick political dividend. But it lacks entirely the distributed administrative capabilities, distributed logistic supply chain, and educative outreach to transform rural populational trend from biomass. Thus, "software" of development – the creation of the correct habit among generations, protection of respiratory health of rural and marginalized population – is critically missing and mega projects are more politically profitable.

Renewable Energy Consumption (% of Total Final Energy Consumption)

This indicator is a measure of the percentage of renewable energy in the state's total final energy use. It fundamentally measures the flexibility of the institutions and political willingness to switch from path dependent fossil fuel industries to sustainable energy markets.

Both states show extreme institutional vulnerabilities in this last indicator, but for very different political-economic reasons. The trend in Bangladesh is characterized by a sharp and sustained decline over the years, from around 60% in 2000 to 25% in 2021 (Appendix, Table 3).

Comparative Energy Infrastructure Trends (%): Bangladesh vs. Japan (2000–2021)

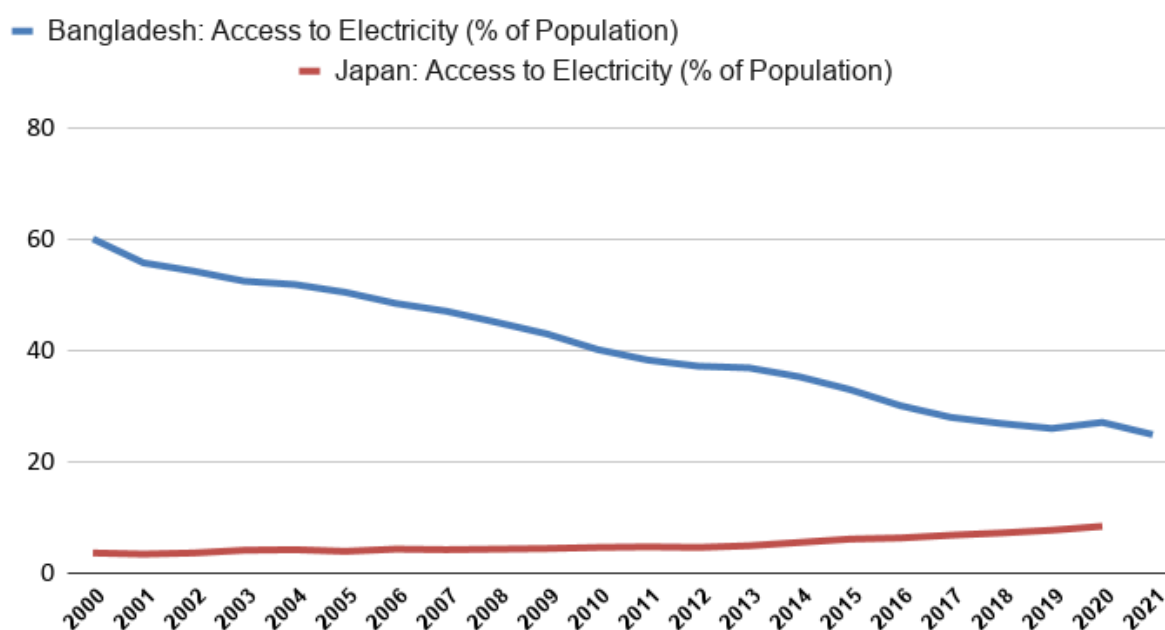


Figure 3: Renewable Energy Consumption (% of total final energy consumption) from 2000 to 2021. Data Source: World Bank.

The tipping point is directly linked to its accelerated electrification drive. In order to reach the 99% grid access at breakneck speed, the state had to heavily depend upon fossil fuels, imported

LNG and coal from international donors. This is an important prioritization of carbon-intensive hardware; it effectively tied the country into long-term fossil fuel deals and stifled and crowded out a modern domestic renewable energy market.

The curve indicates a very slow rate of progress, considering the level of wealth enjoyed by the Japanese people, and even in 2021, it is barely over 8.5%. In 2011, there was a significant turning point after the nuclear disaster of Fukushima Daiichi. This event halted the nation's nuclear fleet and put on the table a historic opportunity for a complete green energy transition. But the power of entrenched path dependency of Japan's "Iron Triangle" intruded. Bureaucrats and utility monopolies soon shifted the state's gears in a heavy direction towards imported fossil fuels, and thus away from the growth of decentralized renewable energy innovation, as they wished to preserve their centralized business models.

Limitations

This comparative MDSD analysis provides valuable insights on the asymmetrical nature of state capacity but has a number of methodological limitations common to comparative empirical studies (Geddes, 2003). The main constraint is that the data are necessarily national averages and are reported by the World Bank, which means that large gaps between regions and between different social and economic groups are inevitably hidden. For example, Bangladesh's overall achievements in grid expansion (close to 100%) obscure localized failures, rolling blackouts, and power quality problems, and reveal that almost all the rural poor still do not have the most basic clean cooking facility they need to live safely. Moreover, there are also micro-level institutional constraints, informal rural economies and other educational gaps which cannot be captured by broad quantitative indicators and which hinder the adoption of regulatory "software" in marginalised communities.

Future Works

Future research is thus needed to move from quantitative to qualitative, localized approaches. Ethnographic fieldwork and qualitative interviews in rural communities could help further the comparison and identify the real cultural and behavioral barriers to adoption of clean cooking. Second, studying in detail the direct effects of democratic backsliding on energy policy, and exploring how strong electoral pressures push LMIC governments toward short-term,

carbon-intensive energy investments and away from long-term sustainability, will add a significant layer to the literature on state capacity. Lastly, a more detailed and analytical study of the path dependency in Japan's Iron Triangle is needed to shed light on the specific legislative and bureaucratic processes that perpetuate the failure to foster sustainable innovation in one of the world's most advanced democracies.

Conclusion

This comparative analysis of SDG 7 has demonstrated that state capacity is not a universal phenomenon and that it is highly asymmetrical, with the need for political survival and institutional path dependency being the most important factors at play. The empirical evidence shows that even in cases of weak states with extreme crisis of electoral legitimacy, like Bangladesh, the state can have a massive, well successful policy outcome by emphasizing the infrastructural 'hardware'. Bangladesh developed a very effective and tested empirical model for other LMICs to address basic energy poverty by combining a dual approach of rapidly expanding fossil fuel grid infrastructure while providing off-grid solar provision.

The longitudinal data at the same time, however, is a cautionary tale: providing politically expedient hardware does not mean that complex regulatory "software" is provided. To achieve universal sustainable development, a lot more is needed than just political will and centralized power plants. It requires local and decentralized governance, extensive behavioral change, and institutional capacity to break through entrenched political-economic interests. Populist survival and donor dependence in South Asia or the bureaucratic inertia of the Iron Triangle in Japan are all examples of the need for a state to change its underlying institutions for true energy transitions to be possible.

References

Acemoglu, D., & Robinson, J. A. (2012). *Why nations fail: The origins of power, prosperity, and poverty*. Crown Publishers.

Anderson, B. (1983). *Imagined communities: Reflections on the origin and spread of nationalism*. Verso.

Bhattacharyya, S. C. (2012). Energy access programmes and sustainable development: A critical review and analysis. *Energy for Sustainable Development*, 16(3), 260-271.

Brubaker, R. (1992). *Citizenship and nationhood in France and Germany*. Harvard University Press.

Dahl, R. A. (1977). On political equality. *Yale Law Journal*, 87(1), 1-15.

Fund for Peace. (2024). *Fragile states index 2024: Country data*.
<https://fragilestatesindex.org/country-data/>

Geddes, B. (2003). *Paradigms and sand castles: Theory building and research design in comparative politics*. University of Michigan Press.

Isumona, V. A. (2012). Imperial presidency and democratic consolidation in Nigeria. *Journal of Democracy*, 23(2), 43-68.

Kaufmann, D., Kraay, A., & Mastruzzi, M. (2010). The worldwide governance indicators: Methodology and analytical issues. *World Bank Policy Research Working Paper No. 5430*.

Khan, A. A. (2023). Caretaker government and democratic backsliding in Bangladesh. *Asian Survey*, 63(2), 245-268.

Lange, M. (2012). State formation and transformation in Africa and Asia. In *The Oxford handbook of transformations of the state* (pp. 115-130). Oxford University Press.

Lijphart, A. (1971). Comparative politics and the comparative method. *American Political Science Review*, 65(3), 682-693.

McCormick, J., Hague, R., & Harrop, M. (2019). *Comparative government and politics: An introduction* (11th ed.). Bloomsbury Publishing.

Müller, J.-W. (2016). *What is populism?* University of Pennsylvania Press.

Newton, K., & van Deth, J. W. (2009). *Foundations of comparative politics* (2nd ed.). Cambridge University Press.

- Pachauri, S., Brew-Hammond, A., Barnes, D. F., Bouille, D. H., Gitonga, S., Modi, V., ... & Zerriffi, H. (2013). Energy access for development. In *Global energy assessment: Toward a sustainable future*. Cambridge University Press.
- Posner, D. N. (2004). The political salience of cultural difference: Why Chewas and Tumbukas are allies in Zambia and adversaries in Malawi. *American Political Science Review*, 98(4), 529-545.
- Rotberg, R. I. (Ed.). (2003). *State failure and state weakness in a time of terror*. Brookings Institution Press.
- Schmitter, P. C., & Karl, T. L. (1991). What democracy is... and is not. *Journal of Democracy*, 2(3), 75-88.
- Skocpol, T. (1985). Bringing the state back in: Strategies of analysis in current research. In P. B. Evans, D. Rueschemeyer, & T. Skocpol (Eds.), *Bringing the state back in* (pp. 3-43). Cambridge University Press.
- Sovacool, B. K., Ali, S. H., Bazilian, M., Radley, B., Nemery, B., Okatz, J., & Mulvaney, D. (2020). Sustainable minerals and metals for a low-carbon future. *Science*, 367(6473), 30-33.
- United Nations. (2023). *The sustainable development goals report 2023: Special edition*. United Nations.
- Varshney, A. (2001). Ethnic conflict and civil society: India and beyond. *World Politics*, 53(3), 362-398.
- Wallerstein, I. (1974). *The modern world-system I: Capitalist agriculture and the origins of the European world-economy in the sixteenth century*. Academic Press.
- World Bank. (2017). *World development report 2017: Governance and the law*. The World Bank.
- World Bank. (2025). *Tracking SDG 7: The energy progress report*. World Bank Group.

Appendix: Data Tables

Table 1: Access to Electricity (% of population)

Source: World Bank, World Development Indicators (2000-2021).

Year	Bangladesh (%)	Japan (%)
2000	32.0	100
2001	35.0	100
2002	37.8	100
2003	40.5	100
2004	40.6	100
2005	44.2	100
2006	50.5	100
2007	46.5	100
2008	54.3	100
2009	57.1	100
2010	55.3	100
2011	59.6	100
2012	65.5	100

Year	Bangladesh (%)	Japan (%)
2013	61.5	100
2014	62.4	100
2015	74.0	100
2016	75.9	100
2017	88.0	100
2018	86.9	100
2019	92.2	100
2020	96.2	100
2021	99.0	100

Table 2: Access to Clean Fuels and Technologies for Cooking (% of population)

Source: World Bank, World Development Indicators (2000-2021).

Year	Bangladesh (%)	Japan (%)
2000	7.7	100

Year	Bangladesh (%)	Japan (%)
2001	8.0	100
2002	8.3	100
2003	8.7	100
2004	9.1	100
2005	9.4	100
2006	10.0	100
2007	10.4	100
2008	11.0	100
2009	11.7	100
2010	12.6	100
2011	13.4	100
2012	14.5	100
2013	15.5	100
2014	16.8	100

Year	Bangladesh (%)	Japan (%)
2015	18.1	100
2016	19.4	100
2017	20.6	100
2018	22.0	100
2019	23.1	100
2020	24.6	100
2021	25.9	100

Table 3: Renewable Energy Consumption (% of total final energy consumption)

Year	Bangladesh (%)	Japan (%)
2000	60.2	3.7
2001	55.9	3.5
2002	54.4	3.7

Year	Bangladesh (%)	Japan (%)
2003	52.6	4.2
2004	52.0	4.3
2005	50.6	4.0
2006	48.6	4.4
2007	47.2	4.3
2008	45.2	4.4
2009	43.1	4.5
2010	40.3	4.7
2011	38.4	4.8
2012	37.3	4.7
2013	37.0	5.0
2014	35.4	5.6
2015	33.1	6.2
2016	30.2	6.4

Year	Bangladesh (%)	Japan (%)
2017	28.1	6.9
2018	27.0	7.3
2019	26.1	7.8
2020	27.2	8.5
2021	25.0	7.0